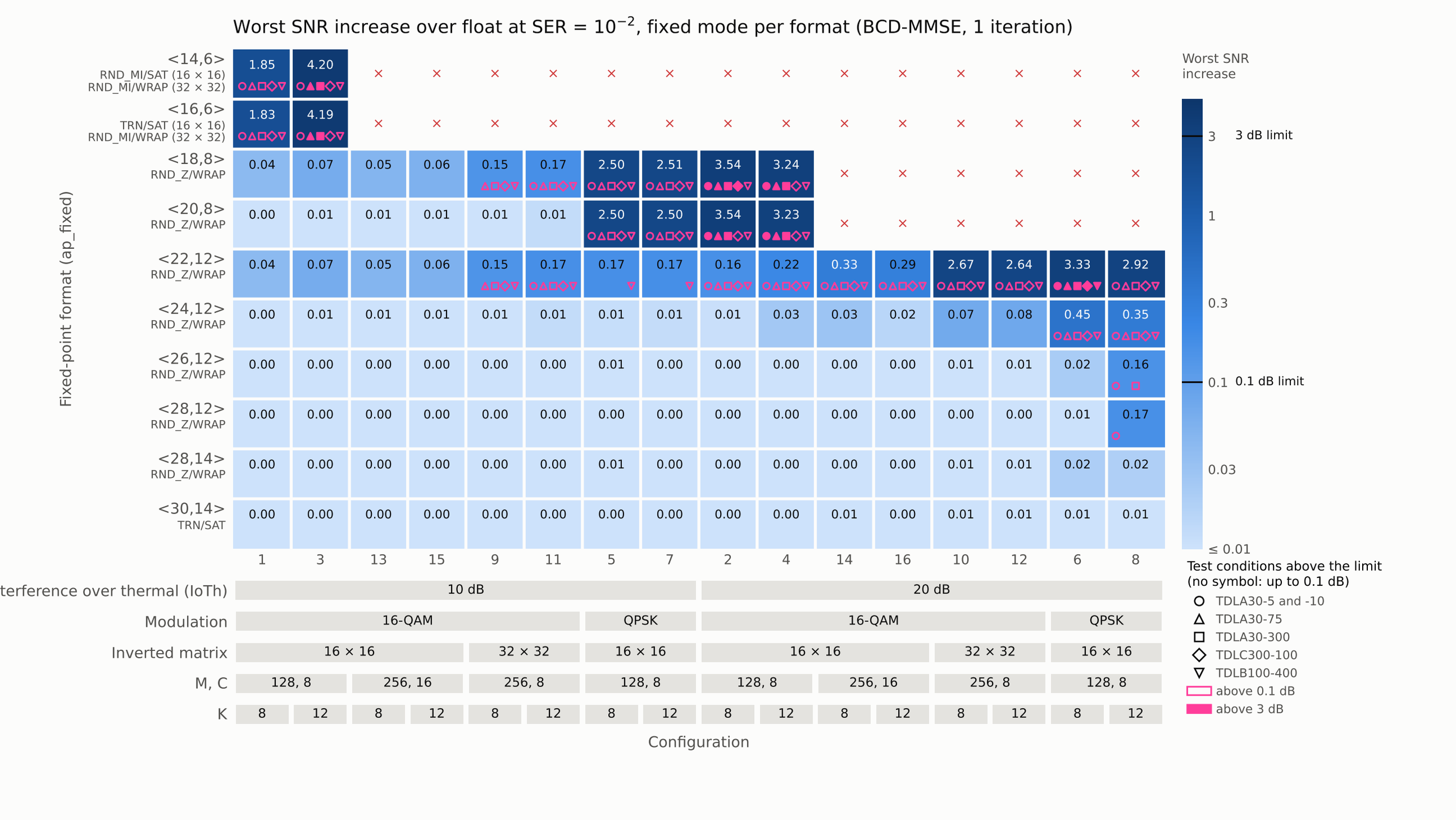
Worst SNR increase over float at SER = 10^{-2}, fixed mode per format (BCD-MMSE, 1 iteration)
Fixed-point format (ap_fixed)
<14,6>
RND_MI/SAT (16 × 16)
RND_MI/WRAP (32 × 32)
1.85 4.20 × × × × × × × × × × × × × ×
<16,6>
TRN/SAT (16 × 16)
RND_MI/WRAP (32 × 32)
1.83 4.19 × × × × × × × × × × × × × ×
<18,8>
RND_Z/WRAP
0.04 0.07 0.05 0.06 0.15 0.17 2.50 2.51 3.54 3.24 × × × × × ×
<20,8>
RND_Z/WRAP
0.00 0.01 0.01 0.01 0.01 0.01 2.50 2.50 3.54 3.23 × × × × × ×
<22,12>
RND_Z/WRAP
0.04 0.07 0.05 0.06 0.15 0.17 0.17 0.17 0.16 0.22 0.33 0.29 2.67 2.64 3.33 2.92
<24,12>
RND_Z/WRAP
0.00 0.01 0.01 0.01 0.01 0.01 0.01 0.01 0.01 0.03 0.03 0.02 0.07 0.08 0.45 0.35
<26,12>
RND_Z/WRAP
0.00 0.00 0.00 0.00 0.00 0.00 0.01 0.00 0.00 0.00 0.00 0.00 0.01 0.01 0.02 0.16
<28,12>
RND_Z/WRAP
0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.01 0.17
<28,14>
RND_Z/WRAP
0.00 0.00 0.00 0.00 0.00 0.00 0.01 0.00 0.00 0.00 0.00 0.00 0.01 0.01 0.02 0.02
<30,14>
TRN/SAT
0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.01 0.00 0.01 0.01 0.01 0.01
1 3 13 15 9 11 5 7 2 4 14 16 10 12 6 8
terference over thermal (IoTh)
10 dB
20 dB
Modulation
16-QAM
QPSK
16-QAM
QPSK
Inverted matrix
16 × 16
32 × 32
16 × 16
16 × 16
32 × 32
16 × 16
M, C
128, 8
256, 16
256, 8
128, 8
128, 8
256, 16
256, 8
128, 8
K
8 12 8 12 8 12 8 12 8 12 8 12 8 12 8 12
Configuration
Worst SNR increase
3
3 dB limit
1
0.3
0.1
0.1 dB limit
0.03
≤ 0.01
Test conditions above the limit
(no symbol: up to 0.1 dB)
TDLA30-5 and -10
TDLA30-75
TDLA30-300
TDLC300-100
TDLB100-400
above 0.1 dB
above 3 dB